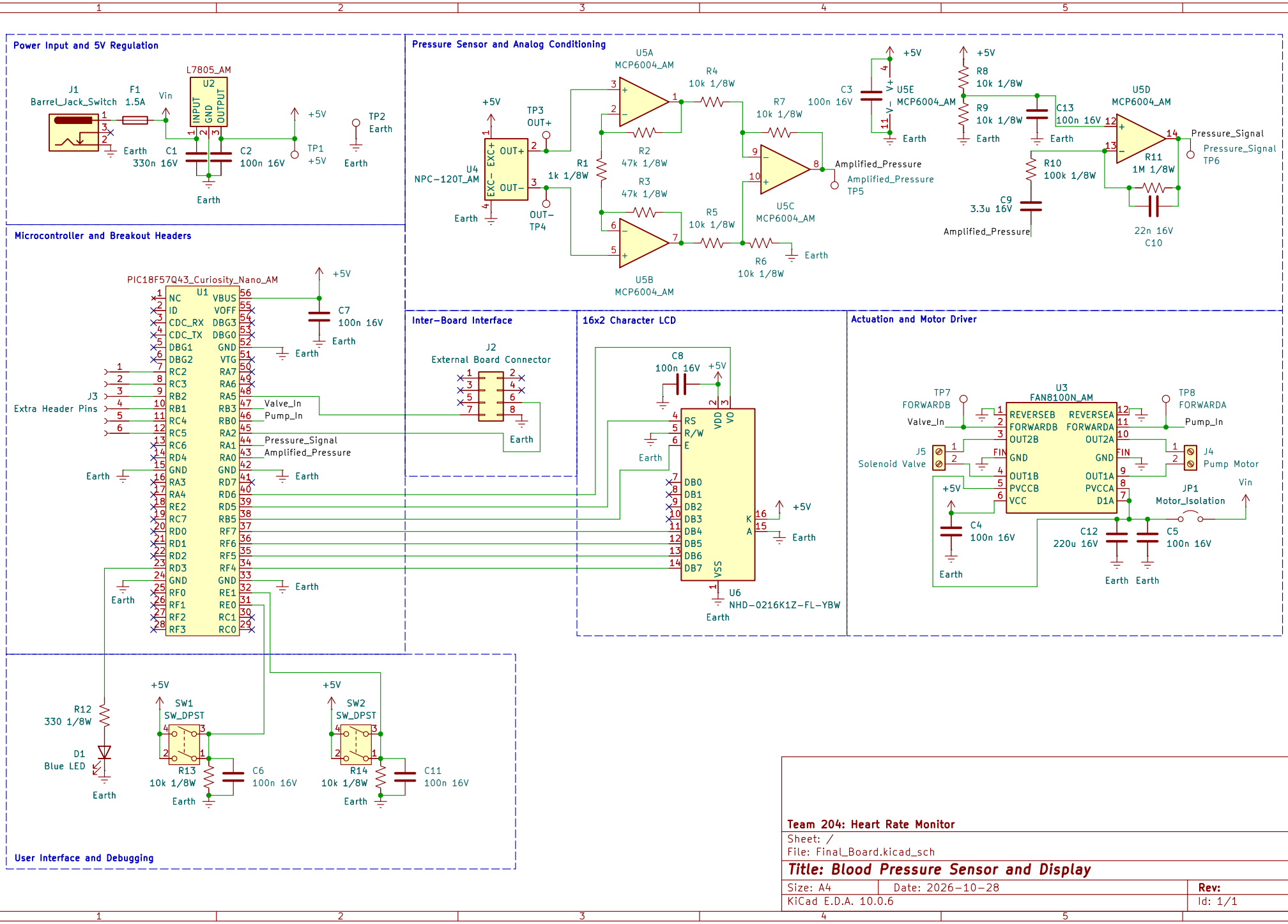

Power Input and 5V Regulation
J1
Barrel_Jack_Switch
F1
1.5A
Vin
L7805_AM
U2
INPUT
GND
OUTPUT
+5V
TP2
Earth
Earth
C1
330n 16V
C2
100n 16V
TP1
+5V
Earth
Pressure Sensor and Analog Conditioning
U5A
MCP6004_AM
+5V
TP3
OUT+
U4
NPC-120T_AM
EXC-
EXC+
OUT+
OUT-
OUT-
TP4
Earth
R1
1k 1/8W
R2
47k 1/8W
R3
47k 1/8W
R4
10k 1/8W
R5
10k 1/8W
R6
10k 1/8W
R7
10k 1/8W
U5B
MCP6004_AM
U5C
MCP6004_AM
U5E
MCP6004_AM
C3
100n 16V
Amplified_Pressure
Amplified_Pressure
TP5
+5V
R8
10k 1/8W
R9
10k 1/8W
C13
100n 16V
U5D
MCP6004_AM
R10
100k 1/8W
C9
3.3u 16V
R11
1M 1/8W
22n 16V
C10
Pressure_Signal
Pressure_Signal
TP6
Microcontroller and Breakout Headers
PIC18F57Q43_Curiosity_Nano_AM
U1
NC
ID
CDC_RX
CDC_TX
DBG1
DBG2
RC2
RC3
RB2
RB1
RC4
RC5
RC6
RD4
GND
RA3
RA4
RE2
RC7
RD0
RD1
RD2
RD3
GND
RF0
RF1
RF2
RF3
VBUS
VOFF
DBG3
DBG0
GND
VTG
RA7
RA6
RA5
RB3
RB0
RA2
RA1
RA0
GND
RD7
RD6
RD5
RB5
RF7
RF6
RF5
RF4
GND
RE1
RE0
RC1
RC0
+5V
C7
100n 16V
Earth
J3
Extra Header Pins
Valve_In
Pump_In
Pressure_Signal
Amplified_Pressure
Earth
Earth
Earth
Inter-Board Interface
J2
External Board Connector
Earth
16x2 Character LCD
C8
100n 16V
+5V
RS
R/W
E
DB0
DB1
DB2
DB3
DB4
DB5
DB6
DB7
VDD
VO
VSS
K
A
+5V
Earth
U6
NHD-0216K1Z-FL-YBW
Earth
Actuation and Motor Driver
TP7
FORWARDB
Valve_In
U3
FAN8100N_AM
REVERSEB
FORWARDB
OUT2B
GND
OUT1B
PVCCB
VCC
REVERSEA
FORWARDA
OUT2A
GND
OUT1A
PVCCA
D1A
FIN
FIN
TP8
FORWARDA
Pump_In
J5
Solenoid Valve
J4
Pump Motor
+5V
C4
100n 16V
Earth
C12
220u 16V
C5
100n 16V
Earth
Earth
JP1
Motor_Isolation
Vin
User Interface and Debugging
R12
330 1/8W
D1
Blue LED
Earth
+5V
SW1
SW_DPST
R13
10k 1/8W
C6
100n 16V
Earth
+5V
SW2
SW_DPST
R14
10k 1/8W
C11
100n 16V
Earth
Team 204: Heart Rate Monitor
Sheet: /
File: Final_Board.kicad_sch
Title: Blood Pressure Sensor and Display
Size: A4
Date: 2026-10-28
Rev:
KiCad E.D.A. 10.0.6
Id: 1/1